Thesis Figure Input Index

This index links current and historical Boundary Element Underground input files to thesis application sections, figure references, and validation evidence.

The Input file column lists runnable DAT files in examples/in/DAT and historical variants retained for traceability. Unavailable historical variants are not runnable inputs.

Figure mappings identify thesis application families only. Consult the Validation or source note column for numerical agreement, discrepancies, and unresolved mappings.

L.D. and L.T. indicate Laplace-domain analysis; T.D. indicates direct time-domain analysis; CQ indicates BDF2 convolution quadrature. The DOMAIN field determines the solver route when a filename or TITLE differs.

Original Fortran sources are in src/fortran-original; validation scripts and comparisons are in validation; solver archives and retained validation reports are in examples/results. Routine PDFs can be regenerated from the retained inputs and archives.

Coverage of current example inputs

All 59 DAT files stored directly in examples/in/DAT are indexed, including the Chapter 3, 5, and 6 cases and refinement variants. Historical files in Prev1992 are cited only as sources.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
1	In-2.1.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.1	Figure 2.1	Source: INFPL1.DAT.
2	In-2.1.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.1	Figure 2.2	Source: INFPL1.DAT.
3	In-2.1.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.1	Figure 2.3	Source: INFPL1.DAT.
4	In-2.1.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.1	Figure 2.4	Source: INFPL1.DAT.
5	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.9	Source: INFPL394.DAT; figure mapping retained.
6	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.10	Source: INFPL394.DAT; figure mapping retained.
7	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.11	Source: INFPL394.DAT; figure mapping retained.
8	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.12	Source: INFPL394.DAT; figure mapping retained.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
9	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.13	Source: INFPL394.DAT; figure mapping retained.
10	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.14	Source: INFPL394.DAT; figure mapping retained.
11	In-2.10.DAT	2D L.D.,SQUARE,INF.,PLANE,0 deg,40 uneq elem,no mean val,val NODE.PLO,disc corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2b	Figure 2.15	Source: INFPL394.DAT; figure mapping retained.
12	In-2.5.DAT	2D L.D.,SQUARE,INF.,PLANE WAVE,0 angle,40 eq elem,no mean val,val at NODE.PLO	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2a	Figure 2.5	Source: INFPL391.DAT.
13	In-2.5.DAT	2D L.D.,SQUARE,INF.,PLANE WAVE,0 angle,40 eq elem,no mean val,val at NODE.PLO	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2a	Figure 2.6	Source: INFPL391.DAT.
14	In-2.5.DAT	2D L.D.,SQUARE,INF.,PLANE WAVE,0 angle,40 eq elem,no mean val,val at NODE.PLO	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2a	Figure 2.7	Source: INFPL391.DAT.
15	In-2.5.DAT	2D L.D.,SQUARE,INF.,PLANE WAVE,0 angle,40 eq elem,no mean val,val at NODE.PLO	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.2a	Figure 2.8	Source: INFPL391.DAT.
16	In-2.16.DAT	2D L.D.,RECT,INF.,PLANE,0 deg,11/9 eq elem,no mean val,val NODE.PLO,cont corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.3	Figure 2.16	Source: IFPLREE1.DAT; current deck uses continuous corners; mapping unconfirmed.
17	In-2.16.DAT	2D L.D.,RECT,INF.,PLANE,0 deg,11/9 eq elem,no mean val,val NODE.PLO,cont corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.3	Figure 2.17	Source: IFPLREE1.DAT; current deck uses continuous corners; mapping unconfirmed.
18	In-2.16.DAT	2D L.D.,RECT,INF.,PLANE,0 deg,11/9 eq elem,no mean val,val NODE.PLO,cont corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.3	Figure 2.18	Source: IFPLREE1.DAT; current deck uses continuous corners; mapping unconfirmed.
19	In-2.16.DAT	2D L.D.,RECT,INF.,PLANE,0 deg,11/9 eq elem,no mean val,val NODE.PLO,cont corn	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.3	Figure 2.19	Source: IFPLREE1.DAT; current deck uses continuous corners; mapping unconfirmed.
20	In-2.20.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.4	Figure 2.20	Source: INFPLSV1.DAT.
21	In-2.20.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.4	Figure 2.21	Source: INFPLSV1.DAT.
22	In-2.20.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.4	Figure 2.22	Source: INFPLSV1.DAT.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
23	In-2.20.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.4	Figure 2.23	Source: INFPLSV1.DAT.
24	In-2.24.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.5	Figure 2.24	Source: INFPLPS1.DAT.
25	In-2.24.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.5	Figure 2.25	Source: INFPLPS1.DAT.
26	In-2.24.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.5	Figure 2.26	Source: INFPLPS1.DAT.
27	In-2.24.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	1. LAPLACE DOMAIN	2.3.1 Plane Wave	2.3.1.5	Figure 2.27	Source: INFPLPS1.DAT.
28	In-2.28.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT WAVE GP 1000 lb bob,NO DECAY, QUBIC ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1a	Figure 2.28	Source: INFEXP1.DAT.
29	In-2.28.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT WAVE GP 1000 lb bob,NO DECAY, QUBIC ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1a	Figure 2.29	Source: INFEXP1.DAT.
30	In-2.28.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT WAVE GP 1000 lb bob,NO DECAY, QUBIC ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1a	Figure 2.30	Source: INFEXP1.DAT.
31	In-2.28.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT WAVE GP 1000 lb bob,NO DECAY, QUBIC ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1a	Figure 2.31	Source: INFEXP1.DAT.
32	In-2.32.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.32	P-wave decay deck; mapping unconfirmed.
33	In-2.32.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.33	P-wave decay deck; mapping unconfirmed.
34	In-2.32.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.34	P-wave decay deck; mapping unconfirmed.
35	In-2.32.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.35	P-wave decay deck; mapping unconfirmed.
36	In-2.36.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P ONLY WAVE GP 1000 lb bob,WITH DECAY,SQUARE ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.36	P-only square-root deck; mapping unconfirmed.

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37	In-2.36.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P ONLY WAVE GP 1000 lb bob,WITH DECAY,SQUARE ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.37	P-only square-root deck; mapping unconfirmed.
38	In-2.36.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P ONLY WAVE GP 1000 lb bob,WITH DECAY,SQUARE ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.38	P-only square-root deck; mapping unconfirmed.
39	In-2.36.DAT	2-D, L.T.,CIRCLE,INFINITY,EXPL POINT P ONLY WAVE GP 1000 lb bob,WITH DECAY,SQUARE ROOT F.	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.1	Figure 2.39	P-only square-root deck; mapping unconfirmed.
40	In-2.40.DAT	PhD Figures 2.40-2.43 reproduction, rectangle P wave at (10,0), with decay	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.2	Figure 2.40	Source: THESIS_RECT_EXPLOSION. DAT.
41	In-2.40.DAT	PhD Figures 2.40-2.43 reproduction, rectangle P wave at (10,0), with decay	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.2	Figure 2.41	Source: THESIS_RECT_EXPLOSION. DAT.
42	In-2.40.DAT	PhD Figures 2.40-2.43 reproduction, rectangle P wave at (10,0), with decay	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.2	Figure 2.42	Source: THESIS_RECT_EXPLOSION. DAT.
43	In-2.40.DAT	PhD Figures 2.40-2.43 reproduction, rectangle P wave at (10,0), with decay	1. LAPLACE DOMAIN	2.3.2 Explosion at a Point	2.3.2.2	Figure 2.43	Source: THESIS_RECT_EXPLOSION. DAT.
44	In-2.44.DAT	2.3.3 Cylindrical cavity in an infinite medium, sudden central internal explosion pressure	1. LAPLACE DOMAIN	2.3.3 Internal explosion	2.3.3	Figure 2.44	Graph agreement not established.
45	In-3.2.DAT	2D, T.D., circle	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.1	Figure 3.2	Source: TINFPL1.DAT.
46	In-3.2.DAT	2D, T.D., circle	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.1	Figure 3.3	Source: TINFPL1.DAT.
47	In-3.2.DAT	2D, T.D., circle	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.1	Figure 3.4	Source: TINFPL1.DAT.
48	In-3.2.DAT	2D, T.D., circle	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.1	Figure 3.5	Source: TINFPL1.DAT.
49	In-3.6.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.2	Figure 3.6	Source: TIFPLSV1.DAT.
50	In-3.6.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.2	Figure 3.7	Source: TIFPLSV1.DAT.
51	In-3.6.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.2	Figure 3.8	Source: TIFPLSV1.DAT.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
52	In-3.6.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.2	Figure 3.9	Source: TIFPLSV1.DAT.
53	In-3.10.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.3	Figure 3.10	Source: TIFPLPS1.DAT.
54	In-3.10.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.3	Figure 3.11	Source: TIFPLPS1.DAT.
55	In-3.10.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.3	Figure 3.12	Source: TIFPLPS1.DAT.
56	In-3.10.DAT	2D L.D., CIRCLE, INFINITY, PLANE WAVE, SV wave	2. TIME DOMAIN	3.3.1 Plane Wave	3.3.1.3	Figure 3.13	Source: TIFPLPS1.DAT.
57	In-3.14.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,NO DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.14	Source: TIFEXP1.DAT; graph agreement not established.
58	In-3.14.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,NO DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.15	Source: TIFEXP1.DAT; graph agreement not established.
59	In-3.14.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,NO DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.16	Source: TIFEXP1.DAT; graph agreement not established.
60	In-3.14.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,NO DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.17	Source: TIFEXP1.DAT; graph agreement not established.
61	In-3.18.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.18	Source: TIFEXP2.DAT; graph agreement not established.
62	In-3.18.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.19	Source: TIFEXP2.DAT; graph agreement not established.
63	In-3.18.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.20	Source: TIFEXP2.DAT; graph agreement not established.
64	In-3.20.DAT	2-D, T.D.,CIRCLE,INFINITY,EXPL POINT P WAVE GP 1000 lb bob,WITH DECAY,	2. TIME DOMAIN	3.3.2 Explosion at a Point	3.3.2.1	Figure 3.20a	Source: TIFEXP2.DAT; historical 3.20a alias absent.
65	In-2.44-time.DAT	2.3.3 Cylindrical cavity in an infinite medium, sudden central internal explosion pressure (time domain)	2. TIME DOMAIN	3.3.3 Internal explosion	3.3.3; see 2.3.3	Figure 2.44	TD counterpart; thesis section 3.3.3 refers to Figure 2.44.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
66	In-3.21-Laplace.DAT	3.21, L.D., HALF-PLANE FREE SURFACE, RAMP VERTICAL PRESSURE	1. LAPLACE DOMAIN	3.3.4 Half-plane internal points	3.3.4(a)	Figures 3.21-3.26	Quantitative mismatch; see validation/In-3.21_thesis_comparison.md.
67	In-3.21-TD.DAT	3.21, T.D., HALF-PLANE FREE SURFACE, RAMP VERTICAL PRESSURE	2. TIME DOMAIN	3.3.4 Half-plane internal points	3.3.4(a)	Figures 3.21-3.26	Quantitative mismatch; see validation/In-3.21_thesis_comparison.md.
68	In-3.27-LD.DAT	3.27, L.D., HALF-PLANE FREE SURFACE, SUDDEN MAINTAINED VERTICAL PRESSURE	1. LAPLACE DOMAIN	3.3.4 Half-plane internal points	3.3.4(b)	Figures 3.27-3.30	Thesis text specifies 0.45 s; Figures 3.29-3.30 show about 0-100 ms.
69	In-3.27-TD.DAT	3.27, T.D., HALF-PLANE FREE SURFACE, SUDDEN MAINTAINED VERTICAL PRESSURE	2. TIME DOMAIN	3.3.4 Half-plane internal points	3.3.4(b)	Figures 3.27-3.30	Thesis text specifies 0.45 s; Figures 3.29-3.30 show about 0-100 ms.
70	In-5.3-LD.DAT	5.3.1.1, L.D., EQUAL-MATERIAL LINED CIRCULAR CAVITY, INFINITE DOMAIN, P WAVE	1. LAPLACE DOMAIN	5.3.1 Lined circular cavity	5.3.1.1	Figures 5.3-5.6 (case family)	Radii 2.50/2.70 m; generic In-5.3.DAT absent.
71	In-5.3-radius5-LD.DAT	5.3.1.1, L.D., FIGURE-COMPARISON 5.0 M CAVITY, EQUAL-MATERIAL LINER	1. LAPLACE DOMAIN	5.3.1 Lined circular cavity	5.3.1.1	Figure 5.3 family	Modified 5.00/5.20 m comparison variant.
72	In-5.3-radius5-TD.DAT	5.3.1.1, T.D., FIGURE-COMPARISON 5.0 M CAVITY, EQUAL-MATERIAL LINER	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.1	Figure 5.3 family	Modified 5.00/5.20 m comparison variant.
73	In-5.3-TD.DAT	5.3.1.1, T.D., EQUAL-MATERIAL LINED CIRCULAR CAVITY, INFINITE DOMAIN, P WAVE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.1	Figures 5.3-5.6 (case family)	Radii 2.50/2.70 m; generic In-5.3.DAT absent.
74	In-5.10a-LD.DAT	5.3.1.2, FIGURE 5.10, L.D., STEEL LINER IN STIFF GRANITE, PLANE P WAVE	1. LAPLACE DOMAIN	5.3.1 Lined circular cavity	5.3.1.2	Figure 5.10	Matched LD/CQ verification; see validation/FIGURE_510_TD_FINDINGS.md.
75	In-5.10b-CQ.DAT	5.3.1.2, FIGURE 5.10, T.D., STEEL LINER IN STIFF GRANITE, PLANE P WAVE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.2	Figure 5.10	Matched LD/CQ verification; see validation/FIGURE_510_TD_FINDINGS.md.
76	In-5.11a-LD.DAT	5.3.1.3, FIGURE 5.11, L.D., STIFF STEEL LINER IN INFINITE SOFT GRANITE, PLANE P WAVE	1. LAPLACE DOMAIN	5.3.1 Lined circular cavity	5.3.1.3	Figure 5.11	Graph agreement not established.
77	In-5.11b-TD.DAT	5.3.1.3, FIGURE 5.11, T.D., STIFF STEEL LINER IN INFINITE SOFT GRANITE, PLANE P WAVE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.3	Figure 5.11	Graph agreement not established.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
78	In-5.13-CQ.DAT	5.3.1.4, FIGURE 5.13, BDF2 C.Q., STIFF STEEL LINER IN INFINITE SOFT GRANITE, EXPLOSION POINT P WAVE	5. BDF2 CONVOLUTION QUADRATURE	5.3.1 Lined circular cavity	5.3.1.4	Figure 5.13	BDF2 CQ route; distinct from direct TD.
79	In-5.13-LD.DAT	5.3.1.4, FIGURE 5.13, L.D., STIFF STEEL LINER IN INFINITE SOFT GRANITE, EXPLOSION POINT P WAVE	1. LAPLACE DOMAIN	5.3.1 Lined circular cavity	5.3.1.4	Figure 5.13	Hoop-stress discrepancy under review.
80	In-5.13-TD.DAT	5.3.1.4, FIGURE 5.13, T.D., STIFF STEEL LINER IN INFINITE SOFT GRANITE, EXPLOSION POINT P WAVE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.4	Figure 5.13	Hoop-stress discrepancy under review.
81	In-5.13-TD-10step.DAT	5.3.1.4, FIGURE 5.13, DIRECT T.D. STEP-SIZE STUDY, 10 STEPS, STEEL LINER IN SOFT GRANITE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.4	Figure 5.13	Step-size variant; see validation/FIGURE_513_STEP_SIZE_STUDY.md.
82	In-5.13-TD-40step.DAT	5.3.1.4, FIGURE 5.13, DIRECT T.D. STEP-SIZE STUDY, 40 STEPS, STEEL LINER IN SOFT GRANITE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.4	Figure 5.13	Step-size variant; see validation/FIGURE_513_STEP_SIZE_STUDY.md.
83	In-5.13-TD-80step.DAT	5.3.1.4, FIGURE 5.13, DIRECT T.D. STEP-SIZE STUDY, 80 STEPS, STEEL LINER IN SOFT GRANITE	2. TIME DOMAIN	5.3.1 Lined circular cavity	5.3.1.4	Figure 5.13	Step-size variant; see validation/FIGURE_513_STEP_SIZE_STUDY.md.
84	In-5.15-LD.DAT	5.3.1.5, FIGURE 5.15, L.D., SOFT CONCRETE LINER IN INFINITE STIFF MEDIUM, EXPLOSION POINT P WAVE	1. LAPLACE DOMAIN	5.3.1 One lined cavity	5.3.1.5	Figure 5.15	Does not reproduce published hoop stresses; see validation/FIGURE_515_FINDINGS.md.
85	In-5.15-TD.DAT	5.3.1.5, FIGURE 5.15, T.D., SOFT CONCRETE LINER IN INFINITE STIFF MEDIUM, EXPLOSION POINT P WAVE	2. TIME DOMAIN	5.3.1 One lined cavity	5.3.1.5	Figure 5.15	Does not reproduce published hoop stresses; see validation/FIGURE_515_FINDINGS.md.
86	In-5.16-LD.DAT	5.3.1.5, FIGURE 5.16, L.D., SOFT CONCRETE LINER IN INFINITE STIFF MEDIUM, EXPLOSION POINT P WAVE	1. LAPLACE DOMAIN	5.3.1 One lined cavity	5.3.1.5	Figure 5.16	Graph discrepancy remains; see validation/FIGURE_516_FINDINGS.md.
87	In-5.16-TD.DAT	5.3.1.5, FIGURE 5.16, T.D., SOFT CONCRETE LINER IN INFINITE STIFF MEDIUM, EXPLOSION POINT P WAVE	2. TIME DOMAIN	5.3.1 One lined cavity	5.3.1.5	Figure 5.16	Graph discrepancy remains; see validation/FIGURE_516_FINDINGS.md.

No.	Input file	Description from DAT TITLE	Program category	Sub-category	Relevant PhD application section	PhD figure reference	Validation or source note
88	In-5.23-CQ.DAT	5.3.1.6A, FIGURE 5.23 HOOP STRESSES, CQ, TWO CONCRETE-LINED CAVITIES IN STIFF GRANITE, VERTICAL P WAVE	5. BDF2 CONVOLUTION QUADRATURE	5.3.1 Two lined cavities	5.3.1.6(a)	Figures 5.23-5.24; mesh 5.22	BDF2 CQ check; see validation/FIGURE_523_LD_TD_FINDINGS.md.
89	In-5.23-LD.DAT	5.3.1.6A, FIGURE 5.23 HOOP STRESSES, LD, TWO CONCRETE-LINED CAVITIES IN STIFF GRANITE, VERTICAL P WAVE	1. LAPLACE DOMAIN	5.3.1 Two lined cavities	5.3.1.6(a)	Figures 5.23-5.24; mesh 5.22	LD/TD differences remain; see validation/FIGURE_523_LD_TD_FINDINGS.md.
90	In-5.23-TD.DAT	5.3.1.6A, FIGURE 5.23 HOOP STRESSES, TD, TWO CONCRETE-LINED CAVITIES IN STIFF GRANITE, VERTICAL P WAVE	2. TIME DOMAIN	5.3.1 Two lined cavities	5.3.1.6(a)	Figures 5.23-5.24; mesh 5.22	LD/TD differences remain; see validation/FIGURE_523_LD_TD_FINDINGS.md.
91	In-5.23-TD-40step.DAT	5.3.1.6A, FIGURE 5.23 HOOP STRESSES, TD, TWO CONCRETE-LINED CAVITIES IN STIFF GRANITE, VERTICAL P WAVE	2. TIME DOMAIN	5.3.1 Two lined cavities	5.3.1.6(a)	Figures 5.23-5.24; mesh 5.22	40-step refinement; see validation/FIGURE_523_LD_TD_FINDINGS.md.
92	In-5.23-TD-80step.DAT	5.3.1.6A, FIGURE 5.23 HOOP STRESSES, TD, TWO CONCRETE-LINED CAVITIES IN STIFF GRANITE, VERTICAL P WAVE	2. TIME DOMAIN	5.3.1 Two lined cavities	5.3.1.6(a)	Figures 5.23-5.24; mesh 5.22	80-step refinement; see validation/FIGURE_523_LD_TD_FINDINGS.md.
93	In-5.25-LD.DAT	5.3.1.6B, FIGURE 5.25 HOOP STRESSES AND FIGURE 5.26 Y-DISPLACEMENTS, LD, CONCRETE LINER (A) AND STEEL LINER (B), VERTICAL P WAVE	1. LAPLACE DOMAIN	5.3.1 Two lined cavities	5.3.1.6(b)	Figures 5.25-5.26; mesh 5.21	Stresses: Figure 5.25; y-displacements: Figure 5.26.
94	In-5.25-TD.DAT	5.3.1.6B, FIGURE 5.25 HOOP STRESSES AND FIGURE 5.26 Y-DISPLACEMENTS, TD, CONCRETE LINER (A) AND STEEL LINER (B), VERTICAL P WAVE	2. TIME DOMAIN	5.3.1 Two lined cavities	5.3.1.6(b)	Figures 5.25-5.26; mesh 5.21	Stresses: Figure 5.25; y-displacements: Figure 5.26.
95	In-5.27-LD.DAT	5.3.2.1, FIGURE 5.17 MESH AND FIGURE 5.18 HOOP STRESSES, LD, STIFF LINER IN SEMI-INFINITE MEDIUM, VERTICAL EXPLOSION P WAVE	1. LAPLACE DOMAIN	5.3.2 Semi-infinite domain	5.3.2.1	Figures 5.17-5.20	LD peaks are close; TD amplitude differs. See validation/FIGURE_527_FINDINGS.md.

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96	In-5.27-TD.DAT	5.3.2.1, FIGURE 5.17 MESH AND FIGURE 5.18 HOOP STRESSES, TD, STIFF LINER IN SEMI-INFINITE MEDIUM, VERTICAL EXPLOSION P WAVE	2. TIME DOMAIN	5.3.2 Semi-infinite domain	5.3.2.1	Figures 5.17-5.20	LD peaks are close; TD amplitude differs. See validation/FIGURE_527_FINDINGS.md.
97	In-5.31-LD.DAT	5.3.2.3, FIGURE 5.27 MESH AND FIGURE 5.31 HOOP STRESSES, LD, TWO STEEL-LINED CAVITIES IN SEMI-INFINITE STIFF GRANITE, VERTICAL P WAVE	1. LAPLACE DOMAIN	5.3.2 Semi-infinite domain	5.3.2.3	Figures 5.31-5.33; mesh 5.27	LD is closer to thesis results; see validation/FIGURE_531_FINDINGS.md.
98	In-5.31-TD.DAT	5.3.2.3, FIGURE 5.27 MESH AND FIGURE 5.31 HOOP STRESSES, TD, TWO STEEL-LINED CAVITIES IN SEMI-INFINITE STIFF GRANITE, VERTICAL P WAVE	2. TIME DOMAIN	5.3.2 Semi-infinite domain	5.3.2.3	Figures 5.31-5.33; mesh 5.27	LD is closer to thesis results; see validation/FIGURE_531_FINDINGS.md.
99	ch6_8_1_1_halfspace_plane_wave.dat	CHAPTER 6.8.1.1 CAVITY IN HALF-SPACE LOADED BY P PLANE WAVE	2. TIME DOMAIN	6.8.1.1 Plane wave	6.8.1.1	Figures 6.18-6.19 (case family)	Direct TD reference deck; graph agreement not established.
100	ch6_8_1_1_halfspace_plane_wave_4elem.dat	CHAPTER 6.8.1.1 HALF-SPACE P WAVE FOUR-ELEMENT VALIDATION	2. TIME DOMAIN	6.8.1 Half-space plane wave	6.8.1.1	Figures 6.18-6.19 (case family)	Reduced 20-step regression variant.
101	In-6.18-LD.DAT	6.8.1.1, FIGURE 6.18 POINT C HOOP STRESS, LAPLACE-DOMAIN HALF-SPACE P-WAVE REFLECTION	1. LAPLACE DOMAIN	6.8.1 Half-space plane wave	6.8.1.1	Figure 6.18	Reflected plane-wave case; graph agreement not established.
102	In-6.18-TD.DAT	6.8.1.1, FIGURE 6.18 POINT C HOOP STRESS, DIRECT TIME-DOMAIN HALF-SPACE P-WAVE REFLECTION	2. TIME DOMAIN	6.8.1 Half-space plane wave	6.8.1.1	Figure 6.18	Reflected plane-wave case; graph agreement not established.
103	In-6.20-LD.DAT	6.8.1.2, FIGURE 6.20 HOOP STRESSES, LD, HALF-SPACE CAVITY, REFLECTED DECAYING EXPLOSION P WAVE	1. LAPLACE DOMAIN	6.8.1 Half-space explosion	6.8.1.2	Figures 6.20-6.21	Reflected explosion case; graph agreement not established.
104	In-6.20-TD.DAT	6.8.1.2, FIGURE 6.20 HOOP STRESSES, TD, HALF-SPACE CAVITY, REFLECTED DECAYING EXPLOSION P WAVE	2. TIME DOMAIN	6.8.1 Half-space explosion	6.8.1.2	Figures 6.20-6.21	Reflected explosion case; graph agreement not established.

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105	In-6.22-LD.DAT	6.8.1.2, FIGURE 6.22 HOOP STRESSES, LD, HALF-SPACE CAVITY, DISCRETIZED GROUND, DECAYING EXPLOSION P WAVE	1. LAPLACE DOMAIN	6.8.1 Half-space explosion	6.8.1.2	Figure 6.22	Discretized-ground explosion case; graph agreement not established.
106	In-6.22-TD.DAT	6.8.1.2, FIGURE 6.22 HOOP STRESSES, TD, HALF-SPACE CAVITY, DISCRETIZED GROUND, DECAYING EXPLOSION P WAVE	2. TIME DOMAIN	6.8.1 Half-space explosion	6.8.1.2	Figure 6.22	Discretized-ground explosion case; graph agreement not established.